

Pitch types are relative

Can MLB's pitch labels be recovered for pitchers a model has never seen, and when the model disagrees, is the model wrong or the label?

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Python · scikit-learn ·
Streamlit

Code: github.com/Andresperez397/pitch-type-classifier

DATA

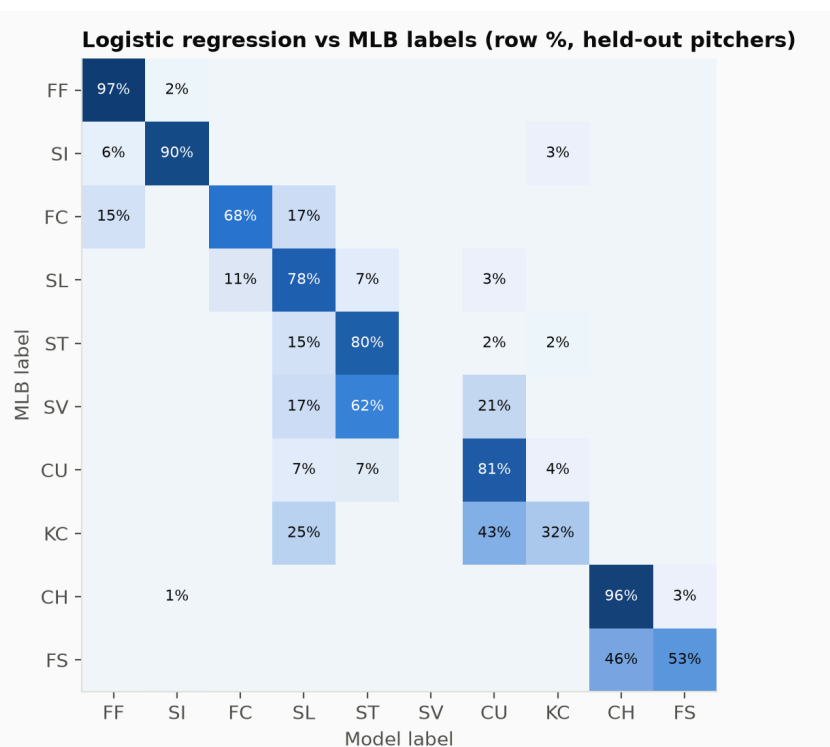
The full 2025 MLB regular season from public Statcast: 712,528 pitches from all 2,430 games, verified against MLB's official schedule. 699,305 pitches from 723 pitchers remain after cleaning. The project rebuilds a 2021 class project with stricter validation.

APPROACH

- **Pre-registered:** labels, features, models and decision rules were committed to git before any model was fit.
- **Unseen pitchers:** 20% of pitchers held out entirely, 5 repeats.
- **Pitcher-relative features:** each pitch is described against the pitcher's own hardest pitches, without using labels.
- **Error audit:** each miss is compared with what other pitchers' nearby pitches are called.

FINDINGS

- **85% accuracy on new pitchers** (macro-F1 0.66; always guessing four-seam: 32%) with a plain logistic regression. Fastballs and changeups work well; knuckle curves and slurve mostly don't.
- **Relative features helped logistic regression in 5 of 5 repeats.** For boosted trees it was 4 of 5, which doesn't meet the pre-declared rule.
- **Flexible models didn't transfer:** boosted trees trailed logistic regression even when tuned on held-out pitchers.
- **About two-thirds of the best model's misses sit where other pitchers' pitches carry the model's label.** The name depends on who throws the pitch.

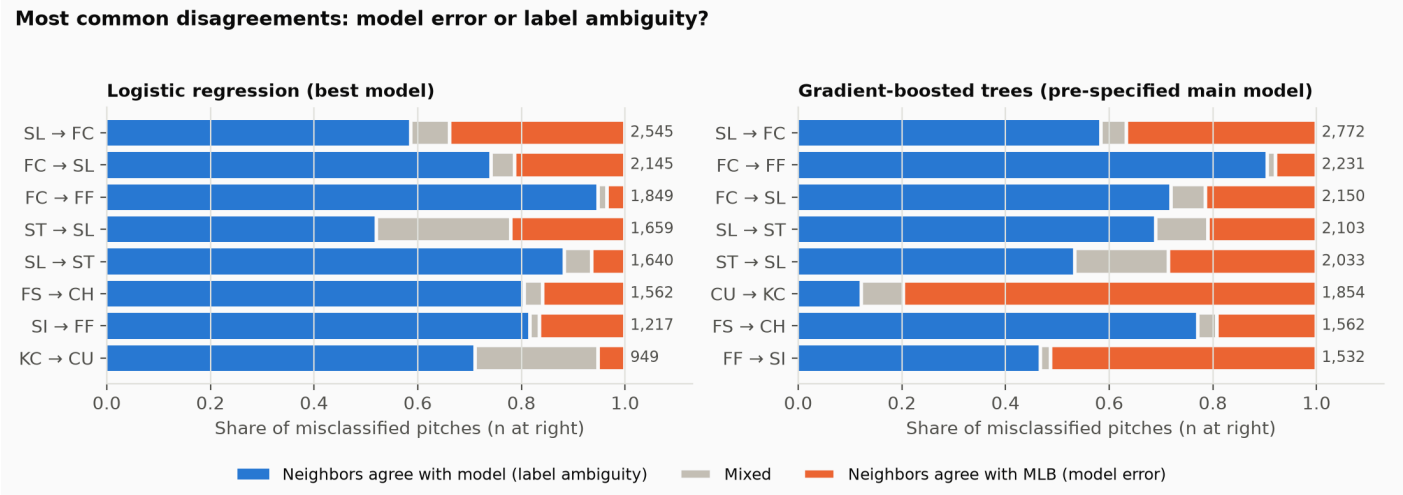


Row-normalized confusion matrix for held-out pitchers (best model). Most confusion sits at naming boundaries: cutter/four-seam, slider/sweeper/cutter, splitter/changeup, knuckle curve/curveball.

MODEL COMPARISON (RELATIVE FEATURES, HELD-OUT PITCHERS)

Model	Accuracy	Macro-F1	Log loss
Logistic regression	0.849	0.660	0.43
Boosted trees, plan's fixed settings	0.807	0.619	2.86
Boosted trees, tuned on held-out pitchers (exploratory)	0.840	0.649	—

Means over 5 repeats; each repeat holds out a different 20% of pitchers.



For each misclassified pitch: do most of its 50 nearest pitches from other pitchers carry MLB's label (model error) or the model's label (label ambiguity)?

INSTRUCTIVE CASES

- **Kenley Jansen:** his signature "cutter" is his fastball, and the model reads it as one.
- **Tyler Rogers (submarine):** his 83 mph sinker drops like a curveball. A population model struggles at extreme arm angles.

LIMITATIONS

MLB's labels are the target, not ground truth. The analysis covers one season and excludes pitchers with fewer than 100 pitches. The hardest-pitch reference assumes the fastest pitches are fastballs. Slurves and knuckle curves are rare classes.

Data: Baseball Savant / Statcast (MLB Advanced Media), public pitch-level data. An interactive arsenal explorer (Streamlit) is included in the repository.

WHY IT MATTERS

Pitch labels are naming decisions, not physical categories. Models that use labels across pitchers (pitch-type splits, pitch-quality models) inherit that inconsistency. Pitcher-relative descriptions are safer.